AA-R-00211H(GL)

April 27, 1977

Used in lieu of
AA-R-211G
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FEDERAL SPECIFICATION

REFRIGERATORS, MECHANICAL, HOUSEHOLD

(ELECTRICAL, SELF-CONTAINED)

This interim Federal specification was developed by the U.S. Army Natick Research and Development Command(GL), Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal specification AA-R-211G.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers self-contained household refrigerators with electric-motor-driven condensing units.

1.2 Classification.

1.2.1 Types, sizes, and grades. Household refrigerators covered by this specification shall be of the following types, sizes, and grades, as specified (see 6.2).

Type I - Low-temperature-food storage compartment below 32 degrees (°) Fahrenheit (F (0°C)) manually initiated defrost with manual or automatic reset.

Type II - Low-temperature-food storage compartment below 32°F (0°C): automatically initiated defrost with automatic reset.

Type III - Frozen-food storage compartment below 12°F (-11°C); manually initiated defrost with automatic reset.

Type IV - Isolated frozen-food storage compartment below 8°F (-13°C); automatically initiated defrost with automatic reset for general-food storage compartment only.

Type V - Isolated frozen-food storage compartment below 8°F (-13°C); automatically initiated defrost with automatic reset for both general-food storage and frozen-food storage compartments.

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Size	Minimum capacity (cu. ft.) (m^3)	Applicable types for size shown
Size 1	1.9 (0.5)	I
Size 2	2.7 (0.8)	I
Size 4	4.0 (0.11)	I, II, III
Size 7	7.0 (0.19)	I, II, III
Size 8	7.6 (0.21)	I, II, III
Size 9	8.6 (0.24)	I, II, III
Size 10	9.5 (0.27)	I, II, III, IV, V
Size 11	10.5 (0.30)	I, II, III, IV, V
Size 12	11.5 (0.32)	I, II, III, IV, V
Size 13	12.5 (0.35)	I, II, III, IV, V
Size 14	13.5 (0.38)	III, IV, V
Size 15	14.5 (0.41)	IV, V
Size 16	15.5 (0.44)	IV, V
Size 17 and over	16.5 (0.47)	IV, V
Size 11	10.5	I, II, III, IV, V

Grade A - Outer shell - Baked enamel on steel.
Inner liner - Aluminum, porcelain enamel, or baked synthetic enamel on steel, or plastic.

Grade B - Outer shell - Porcelain enamel on steel.
Inner liner - Aluminum, porcelain enamel on steel, or plastic.

Grade C - Outer shell - Stainless steel.
Inner liner - Stainless steel.

Grade D - Outer shell - Aluminum.
Inner liner - Aluminum.

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-508	-	Plastic Sheet, Laminated, Decorative, and Nondecorative.
PPP-P-600	-	Porcelain Enamel Products and Household Appliances, Electrical and Mechanical Requirements for Packaging and Packing.

Federal Standard:

FED-STD-123	-	Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-R-12323	-	Refrigerators and Related Equipment, Packaging and Packing Of.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Governmental:

Department of Commerce

CS 35 - Hardwood Plywood.
CS 236 - Mat-Formed Wood Particleboard.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Nongovernmental:

Underwriters Laboratories Inc. (UL)

Standard No. 250 - Household Refrigerators and Freezers (1963).
Standard No. 817 - Cord Sets and Power Supply Cords.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Rd., Melville, L.I., NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050.)

American National Standards Institute, Inc. (ANSI)

B38.1 - Method of Testing for Household Refrigerators, Combination Refrigerator-Freezers, and Household Freezers.

C73.1 - Outlet Receptacles, Attachment Plug Caps and Appliance Plugs.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM)

E 97 - Degree and 0 Degree Directional Reflectance of Opaque Speciemns by Filter Photometery.
B 117 - Salt Spray Test.
C 282 - Method of Test for Acid Resistance of Porcelain Enamel.
C 347 - Reflectivity and Coefficient of Scatter of White Porcelain Enamels.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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U. S. Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. Each type refrigerator delivered under this specification shall be the current standard product of the manufacturer, except for any changes necessary to meet specification requirements.

3.2 Standards.

3.2.1 UL standard. Refrigerators shall conform to the applicable requirements of UL Standard No. 250.

3.2.1.1 UL certification. Prior to approval of the first article, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that refrigerators he proposes to furnish under this specification meet the applicable requirements of UL Standard No. 250. Acceptable evidence of meeting the requirements of this standard shall be the UL Label, UL Listing Mark, or a certified test report from a recognized independent testing laboratory acceptable to the Government indicating that the refrigerators offered have been tested and conform to the applicable requirements of UL Standard No. 250.

3.2.2 Federal Food, Drug, and Cosmetic Act. The interior finish of low-temperature-food and frozen food storage compartments, when baked enamel or acrylic enamel is used shall conform with the applicable requirements of the Federal Food, Drug, and Cosmetic Act (Food Additive resulting from contact with containers or equipment and food additive otherwise affecting food, Section 121.2514-Subpart F, resinous polymeric).

3.2.2.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the refrigerators and all components thereof he proposes to furnish under this specification meet the applicable requirements of the Federal Food, Drug, and Cosmetic Act. Acceptable evidence shall be a certification from a recognized testing laboratory.

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3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his commercial regrigerator provided the completed item complies with all provisions of this specification.

3.3.1 Plastics. All plastics used in the refrigerator shall not impart significant odor or taste to food or water (see 4.3.6.1). Plastics shall not craze, crack, or permanently distort, from temperature, contamination, incompatability with materials of construction or load under conditions of normal use or during storage in temperatures ranging from 0° to 140°F (-18° to 60°C) (see 4.3.6.2.). Plastics shall not show more than a light stain when tested as specified in 4.3.6.3. Clean rework material generated from manufacturing operations may be used by the same manufacturer provided parts produced meet all the applicable requirements of the specification.

3.4 Design. The refrigerators and component parts thereof shall be designed as specified herein. The refrigerators shall be self-contained units comprising an insulated cabinet, evaporator, and electric-motor-driven condensing unit. The interior of the insulated cabinet shall be divided into a general food storage compartment and either a frozen-food or low-temperature-food-storage compartment. The refrigerators shall operate without beaded or running condensation appearing on the exterior of the cabinet (see 4.3.1). Refrigerators with door shelves shall not tip forward when the door shelves are loaded and tested as specified in 4.3.2.

3.4.1 Operating temperatures. The refrigerators shall have the following operating characteristics under no-load conditions (see 4.3.3).

(a) The refrigerators shall be capable of producing average cabinet air temperatures in the general food storage compartment of 36°F (2.2°C) in an ambient of 70°F (21°C), 38°F (3°C) in an ambient of 90°F (32°C), and 41°F (5°C) in an ambient of 110°F (43°C). On 50-Hertz (Hz) electric current operation, the refrigerator shall be capable of producing average cabinet air temperatures in the general food storage compartment of 38°F (3°C) in an ambient of 70°F (21°C), 40°F (4°C) in an ambient of 90°F (32°C) and 45°F (7°C) in ambient of 110°F (43°C).

(b) When operating in the ambient temperatures listed in (a) above, the average cabinet air temperature in the low-temperature-food storage compartment of the types I and II refrigerators shall be lower than 32°F (0°C). The average cabinet air temperature of frozen-food-storage compartments shall be lower than 12°F (-11°C) for the type III refrigerators and lower than 8°F (-13°C) for the types IV and V refrigerators.

3.4.2 Icemaking. Refrigerators shall freeze an amount of water equal to or greater than the capacities of the ice trays standard for the model furnished in 6 hours in an ambient of 90°F (32°C) (see 4.3.4). Refrigerators with automatic icemakers shall be capable of producing and storing, in 24 hours, an amount of ice equal to or greater than the capacity of the ice bucket standard for the model furnished in an ambient of 90°F (32°C) (see 4.3.4).

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3.4.3 Defrosting. Defrosting of each type refrigerator shall be initiated as indicated in 1.2.1. Type II, III, IV, and V refrigerators shall perform according to the operating requirements that follow when tested as specified in 4.3.5. For type II, III, and V, it shall not be necessary to remove frozen food from the low-temperature-food or frozen-food storage compartment during defrosting. The defrosting operation shall not raise the average temperature of frozen food more than 6.5F (-14°C) and the average air temperature of the general food storage compartment more than 10F (-12°C). Allowable temperature increases shall be based on the highest average temperature observed just prior to initiation of defrost. The refrigerator shall not be restored to a cooling operation until the cooling surface is completely defrosted. Normal operation shall resume at any preselected thermostat temperature setting. During a 4-hour recovery period the temperature of the general food storage compartment shall be automatically restored to within 4°F (-15.5°C) of the original stabilized temperature (see 4.3.5.2.2). After the 4-hour recovery period there shall be no accumulation of frost remaining on the cooling surfaces. The development of a light film of white frost is permissible during the recovery period. Re-frozen droplets of water may be adhered to the evaporator, but they shall show signs of having melted during the defrost cycle.

3.4.4 Physical characteristics. The maximum overall dimensions of all sizes of refrigerators and the minimum storage capacities of refrigerators shall be as shown in table I, unless otherwise specified (see 6.2), refrigerators having definite overall dimensions shall be furnished when refrigerators are to be installed in planned locations having space limitations. Storage capacities and shelf areas shall be determined as outlined in ANSI Standard B38-1. The depth or width of the refrigerator shall allow moving the refrigerator through a clear opening 30 inches (0.76 m) wide without removing more than the refrigerator door, hardware, or other readily removable items.

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TABLE I. Refrigerator dimensions and capacities

Size	Maximum overall dimensions Width (in.) (m)	Height (in.) (m)	Depth (in.) (m)	Types	Total storage vol. (cu.ft.) (m^3)	General storage vol. (cu.ft.) (m^3)	Minimum capacities Low-temperature-food frozen-food storage vol. (cu.ft.) (m^3)	Max. Weight (lbs.)
1	20 (0.50)	27 (0.68)	25 (0.63)	I	2.2 (0.66)			150
2	30 (0.76)	25 (0.63)	25 (0.63)	I	2.7 (0.87)			160
4	30 (0.76)	36 (0.91) 2/	27 (0.68)	I, II, III	4.0 (0.11)			180
7	28 (0.71)	60 (1.5)	31 (0.78)	I, II, III	7.0 (0.20) 7.0 (0.20)	6.0 (0.17) 4.0 (0.11)	1/ .80 (0.02)	
8	28 (0.71)	60 (1.5)	31 (0.78)	I, II, III	7.6 (0.21) 7.6 (0.21)	6.5 (0.18) 5.3 (0.15)	1/ .80 (0.02)	
9	28 (0.71)	60 (1.5)	31 (0.78)	I, II, III	8.6 (0.24) 8.6 (0.24)	7.3 (0.20) 6.0 (0.17)	1/ 1.10 (0.03)	
10	28 (0.71)	60 (1.5)	31 (0.78)	I, II, III IV, V	9.5 (0.27) 9.5 (0.27) 9.5 (0.27)	8.1 (0.23) 8.1 (0.23) 6.6 (0.19)	1/ 1.10 (0.03) 1.10 (0.03)	
11	32 (0.81)	64 (1.6)	31 (0.78)	I, II, III, IV, V	10.5 (0.30) 10.5 (0.30)	8.0 (0.23) 6.8 (0.19)	1/ 1.20 (0.03)	
12	32 (0.81)	64 (1.6)	31 (0.78)	I, II, III, IV, V	11.5 (0.32) 11.5 (0.32) 11.5 (0.32)	8.8 (0.25) 7.5 (0.21) 7.5 (0.21)	1/ 1.20 (0.03) 1.9 (0.05)	
13	32 (0.81)	66 (1.7)	31 (0.78)	I, II, III, IV, V	12.5 (0.35) 12.5 (0.35)	8.1 (0.23) 8.1 (0.23)	2.0 (0.6) 2.1 (0.6)	

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TABLE I. Refrigerator dimensions and capacities (cont'd)

Size	Maximum overall dimensions Width (in.) (m)	Height (in.) (m)	Depth (in.) (m)	Types	Total storage vol. (cu.ft.) (m^3)	General storage vol. (cu.ft.) (m^3)	Minimum capacities Low-temperature-food frozen-food storage vol. (cu.ft.) (m^3)	Max. Weight (lbs.)
14	34 (0.9)	69 (1.8)	31 (0.8)	III, IV, V	13.5 (0.38)	8.1 (0.23)	2.1 (0.6)	
15	34 (0.9)	71 (1.8)	31 (0.8)	IV, V	14.5 (0.41)	8.1 (0.23)	3.0 (0.8)	
16	34 (0.9)	71 (1.8)	31 (0.8)	IV, V	15.5 (0.44)	8.1 (0.23)	3.0 (0.8)	
17 and over	48 (1.2	71 (1.8)	31 (0.8)	IV, V	16.5 (0.47)	9.1 (0.26)	5.0 (0.14)	

1/ At least 8 percent of total storage volume.

2/ With finished countertop.

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3.5 Construction.

3.5.1 Cabinets. The design and fabrication of the refrigeration system and cabinet shall prevent water vapor and water from the outside and water from the inside from entering the insulated space during usage and cleaning. The bottom sheet shall be securely fastened to the outer shell and provide a rigid structure to maintain the insulated vapor seal (see 4.3.14). A vapor seal test is not required when foamed-in-place insulation is used exclusively in the cabinet. The underside of carbon steel bottom sheets and the inside surfaces of carbon steel legs and framing of condensing unit compartments of all refrigerators shall be cleaned and treated with the manufacturer's standard corrosion inhibitive methods.

3.5.1.1 Framing. Framing, if used, shall be welded steel or aluminum alloy. No wood shall be used for framing.

3.5.1.2 Outer shells. The outer shells (including the door) of the grades A and B refrigerators shall be carbon-steel sheet, of a thickness not less than 24 gage (0.023 inch nominal (0.6 mm)). The grade A refrigerators shall be finished in baked synthetic enamel. The grade B refrigerators shall be finished in porcelain (vitreous) enamel. Unless otherwise specified (see 6.2), the color of the exteriors of grades A and B refrigerators shall be white. The outer shell of the grade C refrigerators shall be polished stainless steel sheet, of a thickness not less than 24 gage (0.023 inch nominal (0.6 mm)). The outer shell of the grade D refrigerators shall be stainless aluminum-alloy of a thickness not less than 20 gage (0.035 inch nominal (0.9 mm)). If the outer shells of refrigerators are stiffened by channels, braces, or other means, the materials shall be of the following thicknesses: not less than 26 gage (0.017 inch nominal (0.45 mm)) carbon-steel sheet; not less than 26 gage (0.018 inch nominal (0.45 mm)) stainless steel sheet; and not less than 24 gage (0.020 inch nominal (0.6 mm)) stainless aluminum-alloy sheet. The inner surfaces of outer shell of grade A and B refrigerators shall be treated with the manufacturer's standard corrosion inhibitive method. The temperature of the outer shell shall not be more than 10°F (-12°C) below the ambient temperature when the refrigeration system is in operation and the temperature in the storage compartment is being held at the refrigerator's lowest operating range when tested as specified in 4.3.1.

3.5.1.2.1 Porcelain (vitreous) finish. Porcelain (vitreous) enamel finish on metal shall have all coats fused on at a temperature which shall give a satisfactory color and which shall produce a hard, glossy surface. Vitreous enamel or glaze shall be in accordance with commercial practice with respect to discoloration, imperfections, chipped spots, fish scale, burn off, crazing, and blisters which will mar the appearance of the item or leave beads of enamel, glaze, lumps, scumming or the base metal exposed. When vitreous enamel is applied to sheet enameling steel, the visible exposed finish shall consist of at least one ground coat and one finish coat, except for areas where the

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finished coat may be wiped to prevent chippage. Alternate porcelain enamel may be of a type not requiring a ground coat provided a suitable grade of enamel and enameling steel required for this process are used. The finished porcelain enamel shall not exceed 0.018 inch in thickness (see 4.3.7.1). Edges, rims of holes, re-operation areas, and similar locations shall not have a maximum thickness exceeding 0.025 inch (0.6 mm). The reflectance of white porcelain enamel shall be not less than 75 percent (see 4.3.7.2).

3.5.1.2.2 Baked synthetic enamel finishes. Baked synthetic enamel finish shall be applied to phosphatized, zinc-coated steel or to steel, phosphate treated. When exposed to a salt fog for 96 hours, the baked synthetic enamel-coated steel shall show no corrosion (see 4.3.8).

3.5.1.2.3 Finished countertop. When a finished countertop is required for a size 4 refrigerator, the top shall be made of carbon steel or plastic laminate wood, as specified (see 6.2). Carbon steel tops shall have the same color as the cabinet. Wood tops surfaced with laminated plastic sheets shall be minimum 3/4-inch (19 mm) thick and minimum 5-ply plywood or minimum 3/4-inch (19 mm) thick particleboard. Plywood shall be not less than type II, water resistant bond, and face and back veneers not less than sound grade (2) of CS35. Particleboard shall conform to the requirements of type I, grade B, class 1 of CS 236. Decorative laminated plastic sheets for surfacing top of plywood or particleboard core and self-edging shall have a nominal thickness of 1/16-inch (1.6 mm) and conform to style D, type I, grade HP, class 1 of L-P-508. Non-decorative laminated plastic sheets for surfacing the underside of countertops shall have nominal thickness of 1/16-inch (1.6 mm) for balanced construction and conform to style ND, type V, grade HP, of L-P-508. All exposed edges shall be self-edged and all edges eased to avoid sharpness. The pattern, color, and surface finish of decorative laminated plastic sheets shall be as specified (see 6.2).

3.5.1.3 Inner liners of general storage compartments. The inner liner of the grade A refrigerators shall be stainless aluminum-alloy sheet, carbon-steel sheet, or molded plastic. The carbon steel sheet inner liners of the grade A refrigerators shall have the porcelain enamel finish specified in 3.5.1.2.1, or baked synthetic enamel finish specified in 3.5.1.2.2. The inner liner of the grade b refrigerators shall be carbon-steel sheet with the porcelain-enamel finish specified in 3.5.1.2.1 or molded plastic. The inner liner of the grade C refrigerators when specified shall be polished stainless steel sheet (see 6.2). The inner liner of the grade D refrigerators shall be corrosion-resistant aluminum-alloy sheet. Thickness of the sheet metal shall be as specified in 3.5.1.2. The porcelain-enamel finish of the grades A and B refrigerators shall be as specified in 3.5.1.2.1 and, in addition, shall have a class A or better resistance to acid when tested as specified in 4.3.7.3. The unexposed surfaces of carbon steel sheet inner liners shall be treated with the manufacturer's standard corrosion inhibitive method. A sump no less than 1/2 liquid quart (0.0009 m^3) capacity (see 4.3.1.3) shall be provided in the liner floor to accommodate accidental spillage. Unless otherwise specified (see 6.2), the color of the inner liners

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finished with porcelain enamel shall be white or pastel. The molded plastic shall be in accordance with 3.3.1. When plastic liners are used the insulation shall be foamed-in-place polyurethene employing fluorinated hydrocarbons, and the plastic shall be acrylonitrile butadiene styrene (ABS). Plastic liners shall show no cracking or crazing when tested in accordance with 4.3.11. The plastic liner flat surfaces shall have an average thickness of 1/16 inch (1.5 mm) minimum. Corners may go down to a 2-1/32 inch (0.8 mm) minimum thickness.

3.5.1.4 Inner liners of low-temperature-food and frozen-food storage compartment. The inner liners of low-temperature-food and frozen-food storage compartments shall be steel sheet finished with baked synthetic enamel or porcelain enamel, stainless steel sheet, corrosion-resistant aluminum-alloy sheet, corrosion-resistant treated copper-alloy sheet or molded plastic and shall be provided with easily-cleaned corners. When baked enamel or epoxy enamel (Powder painted), on steel is used, the baked enamel or epoxy enamel (Powder painted) shall be in accordance with the manufacturer's standard practice, and in addition shall comply with finish requirements of Federal Food, Drug, and Cosmetic Act (see 3.2.2). When plastic liners are used, the insulation shall be foamed-in-place polyurethane employing fluorinated hydrocarbons and the plastic shall be ABS. Plastic liners shall show no cracking or crazing when tested in accordance with 4.3.11.

3.5.1.5 Exterior doors. Exterior doors shall be of the overlapping type with square, convex-curved, or beveled jambs. Each door shall be provided with an effective thermal barrier between exterior and interior surfaces. Each door shall be equipped with a means for holding it in a closed position with sufficient force to produce an effective gasket seal. A conveniently located handle or hand pull for opening the door shall be provided. If a mechanical latch is used for holding the door closed, it shall be adjustable and capable of being released from inside the refrigerator. Door gaskets shall touch the cabinet evenly and provide an airtight leakproof joint with the door normally closed (see 4.3.9). Reversible doors may be provided. Unless otherwise specified (see 6.2), doors shall swing to the right. A right-hand swing shall be described as having hinges located on the right side of door when facing the refrigerator.

3.5.1.5.1 Inner liners of exterior doors. The inner liners of the exterior doors shall be carbon-steel sheet finished with porcelain enamel specified in 3.5.1.2.1, or baked synthetic enamel specified in 3.5.1.2.2, epoxy enamel (Powder painted), aluminum sheet, stainless steel sheet, or plastic. Plastic door liners when assembled to outer doors, shall not show any cracks or crazing when tested under environmental conditions specified in 4.3.11. When plastic door liners are used and the insulation is foamed-in-place polyurethane employing fluorinated hydrocarbons, the plastic shall be ABS. Unless otherwise specified (see 6.2), the color of the inner liners of doors shall be white or pastel.

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3.5.1.5.2 Breaker strips. Breaker strips made of plastic shall be provided for exterior door openings. Plastic breaker strips shall not show any cracks or crazing when tested under environmental conditions specified in 4.3.11. The bottom breaker strips shall not show any cracks or crazing when tested as specified in 4.3.12. When no breaker strips are provided the general food storage compartment door opening sill shall be tested as specified in 4.3.12. ABS plastic for breaker strips shall be used when foamed-in-place polyurethane insulation with flourinated hydrocarbons is used.

3.5.1.6 Insulation. The space between the outer shells and the inner liners shall be filled with insulation that is resistant to fungus, rot, and vermin. The insulation provided shall be resistant to and installed to prevent settling.

3.5.1.7 Shelving. The general food compartment shall be equipped with shelves of the wire, bar, or reinforced stamped types, except that the shelf over the vegetable drawers or crispers may be of glass or polystyrene plastic. Wire, bar, or stamped shelves shall be made of passivated stainless steel; anodized aluminum alloy; steel plated with copper, nickel, and chromium; steel, hot dipped in tin; or zinc-plated and coated with clear-baked lacquer. Shelves made of metal shall not exhibit more than three visible spots of attack in 12 square inches (0.007 m^2) of shelf area after 24 hours exposure in a salt fog (see 4.3.8). Sliding shelves shall have stops to prevent unintentional complete withdrawal from the refrigerator and to prevent contact damage to the liner when the shelf is pushed into the cabinet.

3.5.1.8 Trays, baffles, crispers, and special storage vessels. Trays, baffles, crispers, and special storage vessels shall be constructed of corrosion-resistant materials.

3.5.1.9 Ice cube trays and grids. Ice cube trays shall be aluminum alloy, stainless steel, polypropylene, or copolymer plastic. Ice cube grids shall be corrosion-resistant. Each metal ice cube tray shall have a mechanical, quick-release mechanism for ejection of ice cubes.

3.5.1.10 Shelf supports. In the event of breakage of a shelf support it shall be replaceable without special tools or removal of storage compartment liner. Shelf supports made of metal shall not exhibit more than three pin-point breakdowns after 96 hours exposure in a salt fog (see 4.3.8).

3.5.1.11 Chromium plating. Brass, aluminum and zinc-alloy, when used in the food zone or for external hardware, shall be chromium plated by first applying a coat of copper and then a coat of nickel. The plating will be uniform, durable, and cover all surfaces. Visible chromium plating on the exterior of the refrigerator cabinets shall be bright or polished.

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3.5.2 Electrical characteristics. Refrigerators shall operate satisfactorily on the electrical system specified. Unless otherwise specified (see 6.2), the refrigerators shall operate on a 120-volt, 60 Hz, single-phase, alternating current electrical system. All refrigerators shall be completely electrically wired. A three-wire cord with a three-prong attachment plug shall be provided and shall extend at least 5 feet, but not more than 9 feet, beyond the point at which it is attached to the back of the cabinet. The cable assembly consisting of a cable and grounding plug connector shall conform to the requirements of UL 817. The cord and attachment plug shall provide grounding of the refrigerator. Refrigerators designed for 50/60 Hz electric current operation shall operate satisfactorily at 50 Hz within plus or minus 10 percent of the manufacturer's design voltage. Any special transformers required to meet specified voltage requirements shall be furnished with each refrigerator. When transformers are required, the transformer capacity shall be 10 percent higher than full load rating of the compressor motor. Unless otherwise specified when 50 Hz operation is specified (see 6.2) a UL approved 3-wire electrical cord shall be provided without a plug. When specified (see 6.2), the item should be supplied with a 3-conductor grounded-type power cord with grounding-type attachment plug in accordance with ANSI C73.1.

3.5.2.1 Lighting. Each refrigerator shall be provided with interior illumination for the general food compartment. A switch for lights shall be operated automatically by opening the door except that when multiple general food compartment doors are supplied, a single pole tumbler-type switch with integral red pilot light may be provided on the front outside of the cabinet in lieu of automatic switches responsive to the opening of either door. Switches for lights which are operated automatically by door openings shall not be located in the bottom breaker strip nor in the bottom outer case panel. Light bulbs shall be protected from condensate drip.

3.5.3 Refrigeration system. The refrigeration system shall be sealed and designed to operate on one of the refrigerants, dichlorodifluoromethane, CCl_2F_2, (R-12) or on monochlorodifluoromethane, $CHClF_2$, (R-22).

3.5.3.1 Condensing unit. The condensing unit shall be an air-cooled type consisting of a motor, compressor, condenser, control, and other necessary equipment.

3.5.3.1.1 Air-cooled condenser. Air-cooled condensers shall be constructed of brass; copper; aluminum; or steel, hot-tin dipped or otherwise suitably protected on the outside after fabrication to resist corrosion. Fins, where used, shall be soldered, welded, or mechanically bonded to the tubing to insure permanent contact. Where the outer shell of the refrigerator or the metal surface of the duct at the rear of the refrigerator is used as an extended condenser cooling surface, the condenser tubes shall be securely bonded to the extended surface by mechanical or other means.

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3.5.3.2 Evaporators. The evaporators shall be either forced-air or gravity type.

3.5.3.2.1 Forced-air evaporators. Forced-air evaporators shall be plate, tube, or finned-tubing type. Finned tubing shall have aluminum tubing with aluminum fins or copper tubing with either copper or aluminum fins. Aluminum fins shall be mechanically bonded to tubing or shall be an integral part of aluminum tubing. When copper tubing with copper fins is used, the entire assembly shall be hot-tin dipped after fabrication. When specified (see 6.2), the evaporator shall be forced air type supplemented by an electric heater for rapid defrosting to eleminate the formation of condensation or frost in the general food storage area.

3.5.3.2.2 Gravity type evaporators. Gravity type evaporators shall be stainless steel; brass or copper plated with tin, nickel, or chromium; aluminum; or galvanized steel.

3.5.3.2.3 Defrost water. Defrost water of all type refrigerators except the isolated frozen food compartment of type IV shall be collected in a readily cleanable, readily removable vessel, or shall be carried out of the cabinet to evaporating apparatus. Means shall be provided to prevent defrost water from dropping on food in the general food storage compartment. Tubes, pipes, or baffles, used to collect or carry defrost water shall have corrosion-resistant surfaces and shall be readily cleanable unless positioned to be self-cleaning by means of condensate entering the device.

3.5.3.3 Refrigerant and oil charge. The assembled refrigeration system shall be completely dehydrated and charged with the proper amount of refrigerant and oil necessary for operation so that no freeze-up occurs during operation (see 4.3.10). Methanol may be used provided its use is approved by the compressor manufacturer and the warranty of the compressor and any other components of the condensing unit is not voided.

3.5.3.4 Refrigeration control. The refrigeration control shall be a temperature control switch activating the compressor motor, fan motor, or both compressor and fan motors. The differential of the temperature control shall be such that the individual temperature changes in the general food storage compartment shall not exceed 6°F (-14°C) and in the low-temperature or frozen-food storage compartments shall not exceed 4°F (-15°C) during no-load cycling operation in accordance with 3.4.1. The temperature control shall be accessible for adjustment and servicing.

3.5.4 Structural soundness. The assembled refrigerator, when completely packed and ready for shipment, shall not be damaged when tested as specified in 4.3.15. Inspection for damage may include all components and shall include inner liners and outer shells; all plastic parts; all interior shelves, trays, baffles, and special storage vessels including their supports; condensing unit; cooling unit; frozen-food or low-temperature-food storage compartment;

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and the refrigerator door. Damage shall include, crazing, cracking, or permanent distortion of plastic and enameled parts; cracking or permanent distortion of metal parts; dislocation of condensing unit, cooling unit, frozen-food storage or low-temperature-food storage compartment, inner liner, or the refrigerator door.

3.6 Marking for identification. The manufacturer's standard nameplate data shall be placed on all refrigerators according to the manufacturer's standard practice.

3.7 Workmanship. Welding and brazing shall be complete, uniform, and properly fused having no burn holes, slag inclusions, scale, or flux deposits and shall not be cracked, fractured, or undercut. Soldering shall be complete, clean, adherent, and without pinholes. Bolts, nuts, screws, studs, and other types of fasteners, when used, shall not be broken, fractured, stripped, or loose and shall have lock washers or shall be a self locking type when used on structural parts subject to vibrations.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements as specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 In-process inspection.

4.2.2.1 In-process examination. Examination shall be for the following to establish that no deviation is made from requirements:

(a) Fastening of bottom sheet as specified in 3.5.1.
(b) Treatment of bottom sheets, legs, and framing of grade A and B refrigerators as specified in 3.5.1.
(c) Application of vitreous enamel as specified in 3.5.1.2.1.

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(d) Unexposed surfaces of carbon steel sheet inner liner treated to resist corrosion as specified in 3.5.1.3.
(e) Use of ABS plastic as specified in 3.5.1.3 and 3.5.1.5.1.
(f) Application of baked synthetic enamel as specified in 3.5.1.2.2.
(g) Installation of insulation as specified in 3.5.1.6.
(h) Application of chromium plating as specified in 3.5.1.11.
(i) Hot-tin dip of copper tubing as specified in 3.5.3.2.1.

4.2.2.2 In-process tests.

4.2.2.2.1 Salt fog test. For the salt fog test in 4.3.8, the lot shall be the material of components of one type offered for inspection at one time. Two specimens shall be used for each test. A specimen shall be one component, or representative sample. Failure of any specimen shall be cause for rejection of the lot.

4.2.2.2.2 Plastics. The plastic tests shall be performed as specified in 4.3.6.1 and 4.3.6.3. Two specimens shall be used for each test. The plastic specimens shall be representative of the plastic used in the refrigerator and shall be taken from the lot of each type of plastic used for refrigerator components offered for inspection at one time. Failure of any specimen shall be cause for rejection of the lot.

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TABLE II. Test schedule

Test characteristics	References: Requirement paragraph	References: Test paragraph	Tests: Inprocess	Tests: Initial production unit	Tests: Production units
Outer shell temp.	3.3, 3.5.1.2	4.3.1		X	
Stability	3.5	4.3.2		X	
No load	3.5.1	4.3.3		X	
Ice making	3.5.2	4.3.4		X	
Defrost	3.5.3	4.3.5		X	
Odor and taste	3.4.1	4.3.6.1	X		
Temperature	3.4.1	4.3.6.2		X	
Stain resistance	3.4.1	4.3.6.3	X		
Porcelain enamel	3.5.1.2.1	4.3.7		X	X
Salt fog	3.5.1.2.2, 3.5.1.3, 3.5.1.7, 3.5.1.10	4.3.8	X		
Door-gasket seal	3.5.1.5	4.3.9		X	X
Refrigeration system moisture	3.5.3.3	4.3.10		X	X
Environmental cracking resistance	3.5.1.5.1 and 3.5.1.5.2	4.3.11		X	
Bottom breaker strip impact	3.5.1.5.2	4.3.12		X	X
Cabinet water tightness	3.5.1	4.3.14		X	
Structural soundness	3.5.4	4.3.15		X	
Electromagnetic compatibility	3.5.2.2	4.3.16		X	

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4.2.3 End item inspection. The inspection lot shall consist of all the refrigerators of one type, size, and grade offered for inspection at one time. The sample unit for this inspection shall be one refrigerator.

4.2.3.1 Visual examination. Examination of the refrigerator shall be in accordance with the classification of defects listed in table III. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE III. Classification of defects

Examine	Defect	Classification Major	Minor
Construction and workmanship, General (applicable to all components and assemblies)	Part missing or not specified type or size (except as otherwise indicated herein).	X	
	Fractured, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, malformed, decayed, or otherwise impaired.	X	
	Defective or otherwise damaged but not affecting serviceability.		X
	Misplaced, loose, or not in proper alignment.	X	
	Sharp burr, sliver, or splinter that may be injurious to personnel.	X	
	Operation omitted or not properly performed.		X
	Functioning component that required abnormal force to operate.	X	
	Not connected or joined as specified or poorly accomplished.	X	
	Adjustable assembly that cannot be adjusted or is not properly adjusted to perform the function intended.	X	
	Component not readily accessible for servicing where required.	X	
	Not finished where required.	X	

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TABLE III. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship, General (applicable to all components and assemblies) (cont'd)	Type of finish not as specified or blistered, peeled, chipped, or area of no film or thin film.	X	
	Color of finish not as specified.		X
	Poor touch up, orange peel, color separation, smudge, fish scale, scratched, stained, overspray, sag imprint, foreign matter imbedded, or finish is otherwise impaired.		X
	Any component or assembly that is painted or otherwise coated where serviceability may be affected.	X	
	Any area of rust.		X
Welding and brazing	Not type specified, missing, incomplete, burn holes, cracked, fractured, or otherwise not fused.	X	
	Slag inclusion, undercut, not smooth and uniform, scale or flux deposits not removed.		X
Soldering (where applicable)	Missing, not adherent, or incomplete.	X	
	Not clean, (flux or flux residue not removed), not smooth, (surface not finished neatly) or pin holes in solder.		X
Bolts, nuts, screws, studs, and other types of threaded fasteners	Missing, not type specified, broken, stripped, fractured, or loose.	X	
Electric assembly	Wiring cut, abraded, not properly joined, loose at terminal, or not inclosed in specified conduit where required.	X	
	Adequate slack not provided for wiring to relieve strain or excessive insulation stripped from wiring.	X	

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TABLE III. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Assembly	Any component not properly assembled or secured.	X	
	Unit out of square or alignment.	X	
Marking, identification	Missing, incomplete, not legible.		X

4.2.3.2 Dimensional examination. Examination shall be made of the end item to determine compliance with dimensional requirements. Any dimension that is not within the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.2.3.3 End item testing. The initial production unit shall be tested in accordance with table II. Any nonconformance shall be cause for rejection of teh unit. The supplier shall perform tests on production units in accordance with 4.3.7, 4.3.9, 4.3.10, and 4.3.12, using an inspection level S-2 and an AQL of 2.5, expressed in terms of defects per hundred units (see table II).

4.2.4 Code and standard compliance. Proof of compliance with the UL requirements of 3.2.1 and the NSF requirements of 3.2.2 shall be submitted to the Government representative.

4.2.5 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected in accordance with the quality assurance provisions of MIL-R-12323.

4.3 Tests.

4.3.1 Outer shell temperature tests. The external surface condensation test shall be performed in accordance with paragraph 7.7 of ANSI Standard B 38.1. The temperature of the outer shell shall then be measured at five locations (front, back, top, and sides) to determine compliance with 3.5.1.2. No beaded or running condensation shall appear on the cabinet or cabinet hardware (see 3.4). A temperature difference of more than 10°F (4°C) between outer shell and ambient; or condensation on the cabinet shall constitute failure of the test.

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4.3.2 Stability. The refrigerator door shelves shall be loaded as follows:

Full width shelves over 3-3/4 inches (95 mm) deep and 10 inches (254 mm) high - 30 pounds per square foot (lbs./sq. ft.) (146 kg/m^2).

Full width shelves less than 3-3/4 inches (95 mm) deep or 10 inches (254 mm) high - 15 lbs./sq. ft (73 kg/m^2).

Butter or cheese compartments - 7 lbs./sq. ft (34 kg/m^2).

Egg shelves - 0.20 lbs. (0.09 kg)/egg.

The remainder of the refrigerator, with all shelves and accessories in place, shall not be loaded. The loaded door shall be opened 90 degrees and the refrigerator observed for tipping to determine compliance with 3.4. Tipping of refrigerator shall constitute failure of the test.

4.3.3 No-load test. To determine compliance with the operating temperature requirements of 3.4.1, a no-load test shall be performed in accordance with the procedure described in ANSI Standard B38.1. Inability to comply with temperature requirements of 3.4.1 shall constitute failure of the test.

4.3.4 Ice-making test. An ice-making test, following the procedure described in ANSI Standard B38.1, shall be performed on the refrigerators to determine compliance with the requirements of 3.4.2. Inability to conform to the ice-making requirements of 3.4.2 shall constitute failure of the test.

4.3.5 Defrost tests. Types II, III, IV, and V refrigerators shall be tested following the procedure described in 4.3.5.1 thru 4.3.5.2.3.2, as applicable to determine compliance with 3.4.3. Inability to conform to the requirements of 3.4.3 shall constitute failure of the test.

4.3.5.1 Defrost test conditions. The frozen-food and low-temperature compartment shall be loaded with test packages and ice trays as described in ANSI Standard B38.1. The refrigerator shall be operated in an ambient of 90°F (32°C) and a relative humidity between 80 and 85 percent. The thermostat shall be adjusted so that the average cabinet air temperature shall be 38°F (3°C), plus or minus 2°F (± 1°C). The refrigerator shall be operated at rated voltage.

4.3.5.2 Defrost test procedure.

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4.3.5.2.1 Door opening schedule. During the first hour of the test, the general food storage compartment door shall be opened wide 15 times for 10-second periods at 4-minute intervals. During an 8-hour period the same procedure shall be repeated during the fourth and seventh hours. The door opening schedule for the type V refrigerator shall be modified so that the frozen-food storage compartment door is opened wide seven times per hour for 10-second periods at 8-minute intervals each time the general food storage compartment door is opened. Prior to the first set of door openings, the general food storage compartment shall be loaded with ice trays filled with 16 to 24 ounces (0.47 to 0.71 L) of 90°F (32°C) water. The number of ice trays shall be determined on the basis of one tray per 2 cubic feet (0.06 m^3) of general food storage volume. A cellulose sponge, approximately 1 inch (25 mm) by 3 inches (76 mm) by 4 inches (102 mm) in saturated size, shall be placed in each tray so the vertical dimension is 3 inches (76 mm). The trays shall be replaced with others containing a sponge and 16 to 24 ounces (0.47 to 0.71 L) of 90°F (32°C) water immediately following the last set of door openings and prior to the 16-hour stabilization period.

4.3.5.2.2 Stabilization. Following the door opening schedule, the refrigerator shall stabilize for the next 16 hours without doors being opened.

4.3.5.2.3 Defrosting. The defrosting of the frozen-food storage compartment shall begin immediately following the stabilization period. The temperature of the general food storage compartment and the frozen-food storage compartment shall be taken continuously, or at intervals of not greater than 15 minutes, starting prior to initiation of the defrost cycles and continuing throughout the recovery period.

4.3.5.2.3.1 Automatically initiated defrost systems (types II, IV, and V refrigerators). Refrigerators with automatically initiated defrost methods may be prepared prior to the test so that they will not defrost during the door opening and stabilization portions of the test. The defrosting period is terminated when the temperature of the general food storage compartment has peaked. The trays and sponges shall then be removed and the refrigerator allowed to operate for a 4-hour recovery period.

4.3.5.2.3.2 Manually initiated defrost systems (type III refrigerators). Type III refrigerators may be opened momentarily to make any defrost baffle adjustments recommended by the manufacturer, and to start the defrost cycle. The defrosting period is terminated for type III when the evaporator surface temperature has peaked. At this time the average air temperature of the general food storage compartment shall be taken. At the termination of the defrost period, any adjustment baffles shall be reset to the same position required to obtain the stabilized cabinet temperature. The defrost water shall be emptied from the collecting trays. The trays and sponges shall be removed from the food compartment. The system shall continue to operate for a 4-hour recovery period.

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4.3.6 Plastic tests.

4.3.6.1 Odor and taste test. Plastics shall be given one of the following odor and taste tests to determine compliance with 3.3.1. Any odor or taste difference shall establish failure of the test.

(a) Place the plastic to be tested with approximately 5-cubic centimeters (cc.) of distilled water in the bottom of a 16-ounce (0.47 L) wide-mouth glass jar. Apat of fresh salted butter, 1-1/4 by 1-1/4 by 1/4 inches (32 by 32 by 6 mm), shall be placed in a shallow dish in the jar, but not in contact with the water or the plastic being tested. The jar shall then be closed tightly. Another pat of butter shall be placed in a second closed glass jar. The two jars shall be kept at 40° to 50°F (4° to 10°C) for 24 hours. At the end of this period, there shall be no significant odor or taste difference between the two pats of butter.

(b) Place the plastic to be tested with approximately 5 cc. of distilled water in a 16-ounce (0.47 L), wide-mouth glass jar. Place a beaker containing 100 to 150 cc. of distilled water in the jar so that the water does not touch the plastic being tested. The jar shall be closed tightly. Place 100 to 150 cc. of distilled water in a second closed glass jar. The two jars shall be kept at 45° to 50F (7° to 10°C) for 24 hours. At the end of this period, there shall be no significant odor or taste difference between the two waters.

4.3.6.2 Temperature tests. Plastic parts of the assembled refrigerator shall be observed under conditions of normal use during conduct of the no-load test (see 4.3.3) to determine compliance with 3.3.1. To determine compliance with plastic storage temperature requirements of 3.3.1, the assembled refrigerator shall be placed in a room maintained at 140°F (60°C) for a 12-hour period. During the last hour of the 12-hour period, the refrigerator shall be subjected to the following two tests (Inability to conform to the requirements of 3.3.1 shall establish failure of the test):

(1) One side of the refrigerator shall be raised 3 inches (76 mm) off a level concrete floor by a single support which can be withdrawn. The refrigerator shall be allowed to drop. This procedure shall be repeated allowing the other three sides to drop.

(2) The refrigerator doors shall be opened 90 degrees and slammed closed 10 times. At the end of the 12-hour period, the refrigerator shall be transferred, as quickly as practical, to a room maintained at 0°F (-18°C). The refrigerator shall remain in this room for a 12-hour period. During the last hour of the 12-hour period, the refrigerator shall be subjected to the two tests (drop and door slamming) described above.

4.3.6.3 Stain resistance test. To determine compliance with the stain resistance requirement of 3.3.1, the following test shall be performed:

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Test specimens shall be placed in contact with tomato juice and shall be partially covered by a watch glass for 15 days at 38°F (3°C). After cleaning according to the manufacturer's recommendations, the plastics shall not show more than a slight stain. Inability to comply with the test requirements of 3.3.1 shall establish failure of the test.

4.3.7 Porcelain (vitreous) enamel tests.

4.3.7.1 Thickness of enamel. The thickness of the porcelain enamel shall be measured in flat portions of the piece at three spots away from edges, reoperation areas, holes, and similar locations subject to normal buildup, with a magnetic thickness gage or other suitable means accurate to plus or minus 0.001 inch (0.02 mm). The average thickness at the three measured spots shall be taken as the thickness of the enamel for the piece to determine compliance with 3.5.1.2.1.

4.4.7.2 Reflectance test. To determine compliance with the requirements of 3.5.1.2.1 the porcelain (vitreous) enamel finish shall be tested for reflectance in accordance with ASTM C347 and ASTM B97. Failure to meet the 75 percent reflectance shall be cause for rejection of the item.

4.3.7.3 Acid resistance test. The test of porcelain enamel for resistance to acid to determine compliance with 3.5.1.3 shall be performed using ASTM C 282.

4.3.8 Saltspray test. Metal components shall be subjected to the salt spray test in accordance with ASTM Standard B 117 to determine compliance with 3.5.1.2.2, 3.5.1.7, and 3.5.1.10.

4.3.9 Door-gasket seal. To determine compliance with 3.5.1.5, the gaskets shall be fitted so that a single strip of paper of the width and thickness of new U.S. paper currency cannot be withdrawn without noticeable resistance of any gasket contact with the door closed. The test shall be performed at least once at each side and at the top and bottom of the door. If the configuration of the door does not permit this test, a 100-watt, minimum, lamp shall be placed inside the refrigerator and an observation for light showing between the gasket and the seating face made in a darkened room. No noticeable resistance to pulling the paper from the gasket and closed door; or visible light between the gasket and the closed door shall establish failure of the test.

4.3.10 Refrigeration system moisture test. The refrigerator shall be operated continuously for 24 hours in an ambient of 110°F (43°C). The power shall then be shut off and the refrigerator allowed to stabilize for 24 hours in the ambient of 70°F (21°C). The refrigerator shall then be operated continuously for 24 hours in the ambient of 70°F (21°C) and observations made for freeze-ups to determine compliance with 3.5.3.3. The temperature control shall be at the lowest setting during both 24-hour periods of operation. A freeze-up shall establish a failure of the test.

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4.3.11 Environmental cracking resistance test. To determine compliance with the exterior door inner liner and the breaker strip environmental cracking resistance requirements of 3.5.1.3, 3.5.1.4, 3.5.1.5.1 and 3.5.1.5.2, the environmental cracking resistance test of ANSI Standard B38.1 shall be performed for a period of 8 days. Any crack or craze shall establish failure of the test.

4.3.12 Bottom breaker strip impact test. To determine compliance with the breaker strip or door-opening sill impact resistance requirement of 3.5.1.5.2, the environmental cracking resistance test of ANSI Standard B38.1 shall be performed. Any crack or craze shall establish failure of the test.

4.3.13 Sump capacity test. To determine conformance with 3.5.1.3, with the inner sump or well empty and dry, a measured 1/2 quart (0.47 L) of tinted liquid shall be poured in the bottom of the general storage compartment to verify the holding camacity provided for spillage. Lesser holding capacity shall constitute failure of this test.

4.3.14 Cabinet water-tightness test. The refrigerator shall be operated continuously for a period of 21 days in an ambient temperature of 90°F (32°C) and a humidity of 85 percent. The internal temperature shall be the lowest setting obtainable. The refrigerator doors shall be closed throughout the test period. At the conclusion of the 21 days of operation, the refrigerator shall be disconnected and allowed to defrost for a period of 6 hours in an ambient temperature of 90°F (32°C) with the doors in a fully open position. After defrosting, all breaker strips shall be removed and an observation made to determine if moisture has accumulated in the insulation or the insulated space to determine compliance with the cabinet water-tightness requirement of 3.5.1. Any evidence of moisture in the insulation shall establish failure of the test.

4.3.15 Structural soundness test. The refrigerator, completely packed at level A, B or C as specified, shall be tested for impact and vibration at the specified A, B, or C test level, respectively, of PPP-P-600 to determine compliance with 3.5.4. Following the test, the refrigerator shall be unpacked, operated, and further examination made for damage. Any evidence of damage shall constitute failure of the test.

5. PACKAGING

5.1 Packaging requirements. The refrigerators shall be preserved-packaged and packed in accordance with the applicable requirements for levels A, B or C as specified in MIL-R-12323 (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

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5.2.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The refrigerators covered by this specification are for home use to preserve food at temperatures above and below freezing while operating in ambient temperatures up to 110°F (43°C). The types III, IV, and V refrigerators are recommended for extended storage of frozen food and ice cream.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type, size, and grade required (see 1.2.1).
(c) Overall dimensions of refrigerators for planned installation (see 3.5.4).
(d) When colors of exteriors and interiors shall be other than specified (see 3.5.1.2, 3.5.1.3, and 3.5.1.5.1).
(e) When only interior liner shall be corrosion-resistant steel (see 3.5.1.3).
(f) Pattern, color, and surface finish of decorated laminated plastic sheets (see 3.5.1.2.3).
(g) Whether doors shall be left hand (see 3.5.1.5).
(h) When other than regular grade of chromium plating is required (see 3.5.1.11).
(i) Electric system operation if other than specified (see 3.5.2).
(j) When 50 Hz operation is required and an ANSI electrical plug is required (see 3.5.2).
(k) When forced air evaporator and heater are required (see 3.5.3.2.1).
(l) Selection of applicable levels of preservation-packaging, and packing (see 5.1).

6.3 Civil agency procurements. When level B preservation and packaging is required for civil agency procurement, the requirements in 3.5.1 of MIL-R-12323, level A preservation and packaging, shall apply.

6.4 Contract data requirements. Any requirements for equipment manuals for the item covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

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AA-R-00211H(GL)

Custodian:

Army - GL

Preparing activity:

Army - GL

Project No. 4110-A341

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NATICK, MASSACHUSETTS 01760

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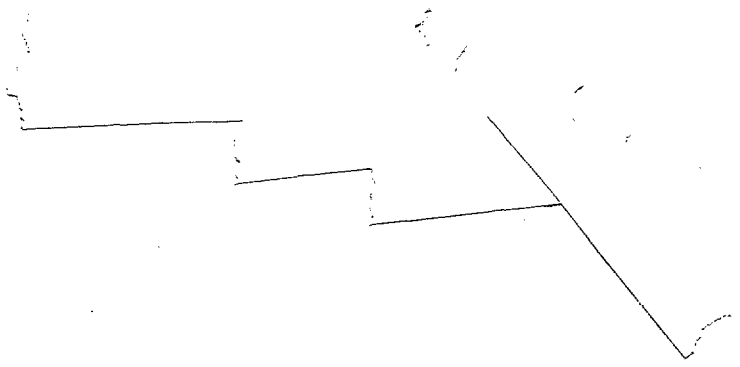
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE Refrigerators, Mechanical, Household (Electrical, Self-Contained) AA-R-00211H(GL)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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